GABRIELLA'S MATH LAB · DECIMALS WORKBOOK

SECTION 1

Decimal Place Value

Every digit has a place. Every place has a value. Learn it, then practice it.

SKILL
One new idea at a time.

SHOW HOW
Watch a problem solved step by step.

TRY ONE
Finish a problem with help.

YOUR TURN
Solve problems on your own.

1.1 The Decimal Point ☐ done 2
1.2 Tenths, Hundredths, and Thousandths ☐ done 4
1.3 Naming the Place of a Digit ☐ done 6
1.4 Finding the Value of a Digit ☐ done 8
1.5 Reading a Place-Value Chart ☐ done 10
1.6 Each Place Is 10 Times the Next ☐ done 12
1.7 Expanded Form ☐ done 14
1.8 From Expanded Form to Standard Form ☐ done 16
1.9 Equivalent Decimals ☐ done 18
✓ Section 1 Review ☐ done 20
🔑 Parent Answer Key 22

Name ______________________________ Started ____________

Georgia K-12 Mathematics Standards 5.NR.4.1, 5.NR.4.2 and 5.NR.1.1. Place-value colors stay the same on every page: **tens**, **ones**, **tenths**, **hundredths**, **thousandths**.

1.1 The Decimal Point

Goal: I can tell the whole-number part from the part less than one.

SHOW HOW

Look at 4.735. What is the whole-number part? Between which two whole numbers is it?

4.735

whole-number part

decimal point

part less than one

① Find the **decimal point**.

② Digits to the **left** are the **whole-number part**: **4**. That is 4 wholes.

③ Digits to the **right** are the **part less than one**: 0.735.

④ 4 wholes plus a part less than one: **4.735 is between 4 and 5**.

Why it works: the decimal point separates whole things (left) from parts of a whole (right). Read it as "and": 4.735 is 4 and 735 thousandths.

LET'S TRY ONE

Look at 12.6.

The whole-number part is _______ . The part less than one is 0._______

So 12.6 is between _______ and _______ .

Name ____________________

YOUR TURN

SKILL 1.1 · THE DECIMAL POINT

1 Write the **whole-number part** of **6.2**.

Answer ____________

2 Write the **whole-number part** of **15.48**.

Answer ____________

3 Write the **part less than one** of **3.9** as a decimal.

Answer ____________

4 Write the **part less than one** of **20.375** as a decimal.

Answer ____________

5 Between which two whole numbers is **7.4**?

Between ____________

6 Between which two whole numbers is **0.86**?

Between ____________

7 Between which two whole numbers is **31.09**?

Between ____________

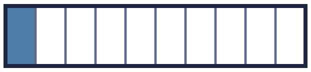
8 Which number is greater than 1: **0.98** or **1.02**? Use the whole-number part to decide.

Greater than 1 ____________

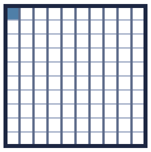
1.2 Tenths, Hundredths, and Thousandths

Goal: I can write the decimal that a model shows.

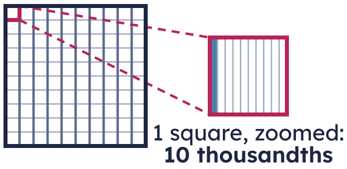
SHOW HOW



10 equal parts: each is **1 tenth = 0.1**

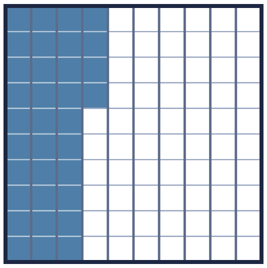


100 squares: each is **1 hundredth = 0.01**



1 square in 10 slices: **1 thousandth = 0.001**

What decimal does this grid show?

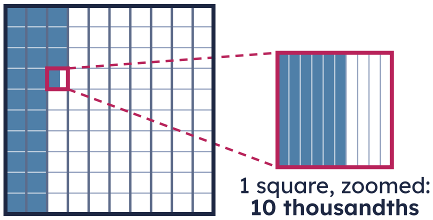


1. The whole grid is **1 whole** cut into 100 equal squares.
2. Count the **full columns**. Each column is 1 tenth: **3 tenths**.
3. Count the **extra squares**. Each is 1 hundredth: **4 hundredths**.
4. Write the digits in their places: **0.34** 34 squares ✓

Why it works: 10 tenths, 100 hundredths, or 1,000 thousandths each make 1 whole. Shading fills one column at a time, top to bottom.

LET'S TRY ONE

What decimal does this grid show? The pink square is cut into 10 equal slices.



_______ full columns = _______ tenths

_______ extra squares = _______ hundredths

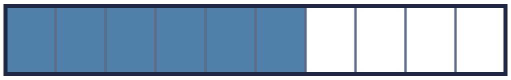
_______ slices = _______ thousandths

Decimal: _______________

YOUR TURN

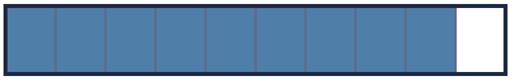
SKILL 1.2 · WRITE THE DECIMAL EACH MODEL SHOWS

1



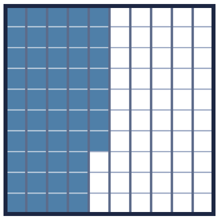
Answer ____________________

2



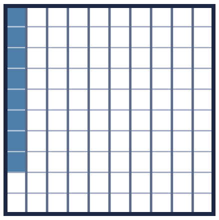
Answer ____________________

3



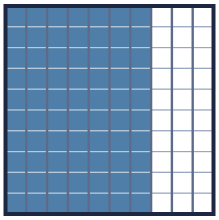
Answer ____________________

4



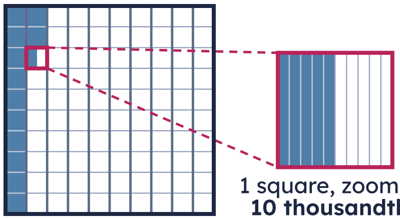
Answer ____________________

5



Answer ____________________

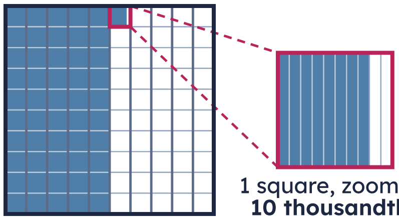
6



1 square, zoomed:
10 thousandths

Answer ____________________

7

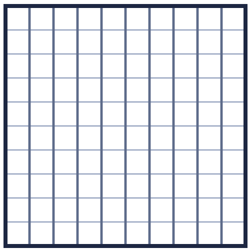


1 square, zoomed:
10 thousandths

Answer ____________________

8

Shade the grid to show **0.62**.



1.3 Naming the Place of a Digit

Goal: I can name the place of any digit in a decimal.

SHOW HOW

In 4.735, what is the place of the 3?

Tens 10	Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
	4	.	7	3	5

1. Write the number in a place-value chart. The decimal point lines up with the chart's point.
2. Count places to the **right** of the point: 1st is **tenths**, 2nd is **hundredths**, 3rd is **thousandths**. To the **left**: 1st is **ones**, 2nd is **tens**.
3. The 3 is the **2nd** digit after the point. It is in the **hundredths** place.
4. **Other way round:** which digit is in the thousandths place? Count 3 places right: the **5**.

Why it works: every place has a name. Places right of the point end in "-ths" because they are parts of one.

LET'S TRY ONE

In 2.618, what is the place of the 8? Fill in the chart first.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
	.			

The 8 is digit number ______ after the point.

The 8 is in the ______________ place.

Name ____________________

YOUR TURN **SKILL 1.3 · NAME THE PLACE OF THE DIGIT**

1 In **5.3**, what is the place of the **3**?

Place ____________________

2 In **0.47**, what is the place of the **7**?

Place ____________________

3 In **8.62**, what is the place of the **8**?

Place ____________________

4 In **0.159**, what is the place of the **9**?

Place ____________________

5 In **26.4**, what is the place of the **2**?

Place ____________________

6 In **3.072**, what is the place of the **7**?

Place ____________________

7 In **14.506**, what is the place of the **6**?

Place ____________________

8 Which digit is in the **hundredths** place of **9.384**?

Digit ____________________

1.4 Finding the Value of a Digit

Goal: I can find how much a digit is worth.

SHOW HOW

What is the value of the 7 in 4.7 35?

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
4	.	7	3	5
4 ones 4		7 tenths 0.7	3 hundredths 0.03	5 thousandths 0.005

1. **Locate** the digit: find the 7 in 4.735.
2. **Identify its place:** the 7 is the 1st digit after the point, the **tenths** place.
3. **Find its value:** 7 tenths = 7 × 0.1 = **0.7**

Why it works: a digit's value depends on its place. The same 7 would be worth 0.07 in the hundredths place, or 7 in the ones place.

LET'S TRY ONE

What is the value of the 6 in 2.169?

The 6 is in the ________________ place.

6 ________________ = ________________

Name ____________________

YOUR TURN **SKILL 1.4 · FIND THE VALUE OF THE DIGIT**

1 What is the value of the **4** in **0.4**?

Value ______________

2 What is the value of the **5** in **3.25**?

Value ______________

3 What is the value of the **7** in **7.81**?

Value ______________

4 What is the value of the **9** in **0.629**?

Value ______________

5 What is the value of the **3** in **5.306**?

Value ______________

6 What is the value of the **7** in **8.075**?

Value ______________

7 What is the value of the **1** in **12.48**?

Value ______________

8 A ribbon is **2.385** meters long. What is the value of the **8**?

Value ______________

1.5 Reading a Place-Value Chart

Goal: I can write the number a chart shows, and put a number into a chart.

SHOW HOW

What number does this chart show?

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
3	.		5	

① Read from the left: **3 ones**. Write **3**.

② The decimal point comes right after the ones: **3.**

③ The **tenths** place is empty. An empty place before the last digit gets a **0**: **3.0**

④ Then **5 hundredths**: **3.05** *not 3.5!*

Why it works: the 0 holds the tenths place so the 5 stays in the hundredths place. Without it, 3.5 would mean 5 tenths.

LET'S TRY ONE

What number does this chart show?

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
0	.	7		2

The empty place is the ________________ place, so it gets a _______ .

The number is ________________

Name ____________________

YOUR TURN **SKILL 1.5 · READ AND FILL PLACE-VALUE CHARTS**

1 What number does the chart show?

Ones 1		Tenths 0.1	Hundredths 0.01
2	.	6	

Number ____________________

2 What number does the chart show?

Ones 1		Tenths 0.1	Hundredths 0.01
0	.	4	9

Number ____________________

3 What number does the chart show?

Ones 1		Tenths 0.1	Hundredths 0.01
5	.	0	8

Number ____________________

4 What number does the chart show?

Ones 1		Tenths 0.1	Hundredths 0.01
1	.		3

Number ____________________

5 What number does the chart show?

Tens 10	Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
4	2	.	1		7

Number ____________________

6 Write **6.904** in the chart.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
	.			

7 Write **0.38** in the chart.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
	.			

8 Write **17.05** in the chart.

Tens 10	Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
		.			

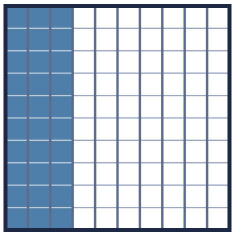
1.6 Each Place Is 10 Times the Next

Goal: I can use “10 of one place make 1 of the place to its left.”

SHOW HOW

1 one = 10 × ← **0.1** tenth = 10 × ← **0.01** hundredth = 10 × ← **0.001** thousandth

How many hundredths are in 3 tenths?



1. 1 tenth is one column. One column has **10** small squares, so **1 tenth = 10 hundredths.**
2. 3 tenths = 3 × 10 hundredths = **30 hundredths.** 0.3 = 0.30 ✓
3. **Two places:** 1 tenth = 10 hundredths = **100 thousandths** (10 × 10).
4. **Other way:** group by 10s. 40 hundredths = 4 groups of 10 hundredths = **4 tenths.**

How many times as much? The 4 in **0.4** is 1 place left of the 4 in **0.04**, so it is worth **10 times** as much. 2 places left: 10 × 10 = **100 times**. Going right, 1 place is worth **one tenth** as much: 0.04 is $^{1}/_{10}$ of 0.4.

LET'S TRY ONE

How many thousandths are in 5 hundredths?

1 hundredth = ________ thousandths

5 hundredths = 5 × ________ = ________ thousandths

Name ____________________

YOUR TURN

SKILL 1.6 · 10 TIMES RELATIONSHIPS

1 1 tenth = _______ hundredths

2 1 hundredth = _______ thousandths

3 6 tenths = _______ hundredths

4 1 tenth = _______ thousandths

5 40 hundredths = _______ tenths

6 The value of the 7 in **0.7** is how many times the value of the 7 in **0.07**?

Answer ________________ times

7 The value of the 5 in **0.5** is how many times the value of the 5 in **0.005**?

Answer ________________ times

8 The 3 in **3.4** is worth how many times the 3 in **0.36**?

Answer ________________ times

1.7 Expanded Form

Goal: I can write a decimal as the sum of its place values.

SHOW HOW

Write 4.735 in expanded form.

Ones 1		**Tenths** 0.1	**Hundredths** 0.01	**Thousandths** 0.001
4	.	**7**	**3**	**5**
4 ones **4**		7 tenths **0.7**	3 hundredths **0.03**	5 thousandths **0.005**

1. Find the **value of each digit** (Skill 1.4): 4, 0.7, 0.03, 0.005.
2. Write the values with **+** between them, largest place first.
3. A **0 digit** adds nothing, so skip it: 6.208 = 6 + 0.2 + 0.008.

4 + 0.7 + 0.03 + 0.005

Why it works: a decimal is the total of its digits' values. Expanded form shows each part. Standard form (4.735) is the short way to write the total.

LET'S TRY ONE

Write 3.62 in expanded form.

3 ones = ________ 6 tenths = ________ 2 hundredths = ________

3.62 = ________ + ________ + ________

Name ____________________

YOUR TURN SKILL 1.7 · WRITE EACH NUMBER IN EXPANDED FORM

1 Write **0.85** in expanded form.

= ____________________

2 Write **2.5** in expanded form.

= ____________________

3 Write **0.46** in expanded form.

= ____________________

4 Write **5.13** in expanded form.

= ____________________

5 Write **7.042** in expanded form.

= ____________________

6 Write **0.309** in expanded form.

= ____________________

7 Write **15.27** in expanded form.

= ____________________

8 Write **30.605** in expanded form.

= ____________________

1.8 From Expanded Form to Standard Form

Goal: I can write the standard form of a number given in expanded form.

SHOW HOW

Write 5 + 0.2 + 0.009 in standard form.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
5	.	2	0	9

1. Put each value's digit in its place: **5** ones, **2** tenths, **9** thousandths.
2. No hundredths are listed, so write a **0** in the hundredths place (highlighted).
3. Read the chart: **5.209** *not 5.29!*

Why it works: expanded form lists the value in each place. Standard form writes each digit in its place, with 0 in any empty place before the last digit.

LET'S TRY ONE

Write 0.6 + 0.04 + 0.003 in standard form.

Ones 1		Tenths 0.1	Hundredths 0.01	Thousandths 0.001
	.			

Standard form: ________________

Name ____________________

YOUR TURN

SKILL 1.8 · WRITE EACH NUMBER IN STANDARD FORM

1 2 + 0.4

Standard form ____________________

2 0.7 + 0.05

Standard form ____________________

3 8 + 0.3 + 0.06

Standard form ____________________

4 4 + 0.08

Standard form ____________________

5 0.9 + 0.001

Standard form ____________________

6 0.03 + 0.5 (careful: not in order)

Standard form ____________________

7 10 + 6 + 0.2 + 0.007

Standard form ____________________

8 0.004 + 3 + 0.1

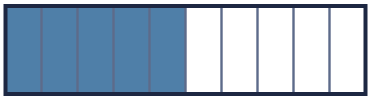
Standard form ____________________

1.9 Equivalent Decimals

Goal: I can tell when two decimals name the same amount.

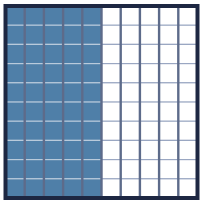
SHOW HOW

Is 0.5 equal to 0.50?



0.5 = 5 tenths

=



0.50 = 50 hundredths

1. 0.5 is **5 tenths**. 1 tenth = 10 hundredths, so 5 tenths = **50 hundredths** = 0.50.
2. The same amount is shaded. **Yes: 0.5 = 0.50 = 0.500.**
3. **Watch out:** a zero right after the point is different. 0.05 is 5 hundredths, **not** equal to 0.5.

Why it works: a zero added at the **end**, after the last decimal digit, does not move any digit, so the value stays the same.

LET'S TRY ONE

Is 0.7 equal to 0.70?

0.7 = 7 tenths = ________ hundredths. 0.70 = ________ hundredths.

Equal? (yes or no) ________________

YOUR TURN **SKILL 1.9 · ARE THEY EQUAL? WRITE YES OR NO.**

1 **0.3** and **0.30**

Yes or no ____________

2 **0.6** and **0.06**

Yes or no ____________

3 **1.2** and **1.20**

Yes or no ____________

4 **0.40** and **0.4**

Yes or no ____________

5 **2.5** and **2.05**

Yes or no ____________

6 **0.700** and **0.7**

Yes or no ____________

7 Write a decimal with **hundredths** that is equal to **0.9**.

Decimal ____________

8 Write a decimal with **thousandths** that is equal to **3.2**.

Decimal ____________

✓ Section 1 Review

Mixed practice from Skills 1.1 to 1.9. No hints: show what you know.

YOUR TURN **REVIEW · PROBLEMS 1–6**

1 In **7.164**, what is the place of the **4**?

Place ____________

2 A beetle is **0.075** meter long. What is the value of the **7**?

Value ____________

3 Write **0.582** in expanded form.

= ____________

4 Write **6 + 0.04 + 0.009** in standard form.

Standard form ____________

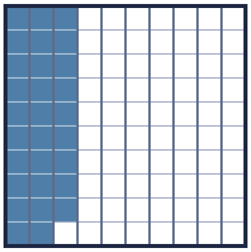
5 Is **0.80** equal to **0.8**?

Yes or no ____________

6 2 tenths = ______ hundredths

YOUR TURN **REVIEW · PROBLEMS 7–10**

7 What decimal does the grid show?



Answer ____________________

8 Which digit is in the **thousandths** place of **4.317**?

Digit ____________________

9 Between which two whole numbers is **18.93**?

Between ____________________

10 The value of the 9 in **0.9** is how many times the value of the 9 in **0.009**?

Answer ____________________ times

HOW DID IT GO?

Circle the skills to practice again, then redo two problems from that skill's Your Turn page. **1.1** **1.2** **1.3** **1.4** **1.5** **1.6** **1.7** **1.8** **1.9**

Parent Answer Key

Equal values are correct: 0.7 and 0.70 are the same answer. Expanded form may also be written $4 \times 1 + 7 \times 0.1$. “Try” is the Let’s Try One problem.

1.1 The Decimal Point

Try **12 and 13** Whole-number part 12; part less than one 0.6; so 12.6 is between 12 and 13.

1 **6** The digits left of the decimal point are the whole-number part: 6.

2 **15** The digits left of the decimal point are the whole-number part: 15.

3 **0.9** The digits right of the decimal point are the part less than one: 0.9.

4 **0.375** The digits right of the decimal point are the part less than one: 0.375.

5 **7 and 8** 7.4 is 7 wholes and a part less than one, so it is between 7 and 8.

6 **0 and 1** 0.86 is 0 wholes and a part less than one, so it is between 0 and 1.

7 **31 and 32** 31.09 is 31 wholes and a part less than one, so it is between 31 and 32.

8 **1.02** 1.02 has whole-number part 1, so it is at least 1. 0.98 has whole-number part 0, so it is less than 1.

1.2 Tenths, Hundredths, and Thousandths

Try **0.236** 2 full columns = 2 tenths, 3 extra squares = 3 hundredths, 6 slices = 6 thousandths.

1 **0.6** 6 of 10 equal parts shaded = 6 tenths.

2 **0.9** 9 of 10 equal parts shaded = 9 tenths.

3 **0.47** 4 full columns = 4 tenths, 7 extra squares = 7 hundredths.

4 **0.08** 0 full columns = 0 tenths, 8 extra squares = 8 hundredths.

5 **0.7** 7 full columns = 7 tenths = 0.7. (0.70 is also correct.)

6 **0.125** 1 full column = 1 tenth, 2 extra squares = 2 hundredths, 5 slices = 5 thousandths.

7 **0.508** 5 full columns = 5 tenths, 0 extra squares = 0 hundredths, 8 slices = 8 thousandths.

8 **62 squares: 6 full columns and 2 more squares** 0.62 = 6 tenths (6 columns) + 2 hundredths (2 squares).

1.3 Naming the Place of a Digit

Try **thousandths** Chart: 2 ones, 6 tenths, 1 hundredth, 8 thousandths. The 8 is digit number 3 after the point: the thousandths place.

1 **tenths** In 5.3, the 3 is in the tenths place.

2 **hundredths** In 0.47, the 7 is in the hundredths place.

3 **ones** In 8.62, the 8 is in the ones place.

4 **thousandths** In 0.159, the 9 is in the thousandths place.

5 **tens** In 26.4, the 2 is in the tens place.

6 **hundredths** In 3.072, the 7 is in the hundredths place.

7 **thousandths** In 14.506, the 6 is in the thousandths place.

8 **8** The hundredths digit of 9.384 is 8.

1.4 Finding the Value of a Digit

Try **0.06** The 6 is in the hundredths place: 6 hundredths = 0.06.

1 **0.4** The 4 is in the tenths place: 4 tenths = 0.4.

2 **0.05** The 5 is in the hundredths place: 5 hundredths = 0.05.

3 **7** The 7 is in the ones place: 7 ones = 7.

4 **0.009** The 9 is in the thousandths place: 9 thousandths = 0.009.

5 **0.3** The 3 is in the tenths place: 3 tenths = 0.3.

6 **0.07** The 7 is in the hundredths place: 7 hundredths = 0.07.

7 **10** The 1 is in the tens place: 1 ten = 10.

8 **0.08** The 8 is in the hundredths place: 8 hundredths = 0.08.

1.5 Reading a Place-Value Chart

Try **0.702** The empty place is the hundredths place, so it gets a 0: 0.702.

1 **2.6** 2 ones, 6 tenths → 2.6.

2 **0.49** 0 ones, 4 tenths, 9 hundredths → 0.49.

3 **5.08** 5 ones, 0 tenths, 8 hundredths → 5.08.

4 **1.03** 1 one, 3 hundredths; the empty tenths place gets a 0 → 1.03.

5 **42.107** 4 tens, 2 ones, 1 tenth, 7 thousandths; the empty hundredths place gets a 0 → 42.107.

6 **6 ones, 9 tenths, 0 hundredths, 4 thousandths** 6.904: the 0 must be written in the hundredths place.

7 **0 ones, 3 tenths, 8 hundredths** The thousandths box may be left empty or hold a 0 (0.380 = 0.38).

8 **1 ten, 7 ones, 0 tenths, 5 hundredths** 17.05: the 0 holds the tenths place.

1.6 Each Place Is 10 Times the Next

Try **50** Each smaller place needs 10 times as many: 5 hundredths = 50 thousandths.

1 **10** Each smaller place needs 10 times as many: 1 tenth = 10 hundredths.

2 **10** Each smaller place needs 10 times as many: 1 hundredth = 10 thousandths.

3 **60** Each smaller place needs 10 times as many: 6 tenths = 60 hundredths.

4 **100** Each smaller place needs 10 times as many: 1 tenth = 10 hundredths = 100 thousandths.

5 **4** Group by 10s: 40 hundredths = 4 groups of 10 hundredths = 4 tenths.

6 **10** tenths vs hundredths: the digit is 1 place to the left, so 10 times as much.

7 **100** tenths vs thousandths: the digit is 2 places to the left, so 10 × 10 = 100 times as much.

8 **10** ones vs tenths: the digit is 1 place to the left, so 10 times as much.

1.7 Expanded Form

Try **3 + 0.6 + 0.02** 3 ones = 3, 6 tenths = 0.6, 2 hundredths = 0.02.

1 **0.8 + 0.05** 8 tenths, 5 hundredths.

2 **2 + 0.5** 2 ones, 5 tenths.

3 **0.4 + 0.06** 4 tenths, 6 hundredths.

4 **5 + 0.1 + 0.03** 5 ones, 1 tenth, 3 hundredths.

5 **7 + 0.04 + 0.002** 7 ones, 4 hundredths, 2 thousandths (the 0 in the tenths place adds nothing).

6 **0.3 + 0.009** 3 tenths, 9 thousandths (the 0 in the hundredths place adds nothing).

7 **10 + 5 + 0.2 + 0.07** 1 ten, 5 ones, 2 tenths, 7 hundredths.

8 **30 + 0.6 + 0.005** 3 tens, 6 tenths, 5 thousandths (the 0s in the ones and hundredths places add nothing).

1.8 From Expanded Form to Standard Form

Try **0.643** Put each value in its place: 0.6 + 0.04 + 0.003 = 0.643.

1 **2.4** Put each value in its place: 2 + 0.4 = 2.4.

2 **0.75** Put each value in its place: 0.7 + 0.05 = 0.75.

3 **8.36** Put each value in its place: 8 + 0.3 + 0.06 = 8.36.

4 **4.08** Put each value in its place: 4 + 0.08 = 4.08.

5 **0.901** Put each value in its place: 0.9 + 0.001 = 0.901.

6 **0.53** Put each value in its place: 0.03 + 0.5 = 0.53.

7 **16.207** Put each value in its place: 10 + 6 + 0.2 + 0.007 = 16.207.

8 **3.104** Put each value in its place: 0.004 + 3 + 0.1 = 3.104.

1.9 Equivalent Decimals

Try **yes** 0.7 = 7 tenths = 70 hundredths; 0.70 = 70 hundredths; so yes.

1 **yes** 0.3 = 0.30: zeros added after the last decimal digit do not change the value.

2 **no** 0.6 ≠ 0.06: the digits are in different places.

3 **yes** 1.2 = 1.20: zeros added after the last decimal digit do not change the value.

4 **yes** 0.40 = 0.4: zeros added after the last decimal digit do not change the value.

5 **no** 2.5 ≠ 2.05: the digits are in different places.

6 **yes** 0.700 = 0.7: zeros added after the last decimal digit do not change the value.

7 **0.90** 9 tenths = 90 hundredths.

8 **3.200** Adding zeros after the last decimal digit keeps the value: 3.2 = 3.20 = 3.200.

Review Section 1 Review

1 **thousandths** In 7.164, the 4 is in the thousandths place.

2 **0.07** The 7 is in the hundredths place: 7 hundredths = 0.07.

3 **0.5 + 0.08 + 0.002** 5 tenths, 8 hundredths, 2 thousandths.

4 **6.049** Put each value in its place: 6 + 0.04 + 0.009 = 6.049.

5 **yes** 0.80 = 0.8: zeros added after the last decimal digit do not change the value.

6 **20** Each smaller place needs 10 times as many: 2 tenths = 20 hundredths.

7 **0.29** 2 full columns = 2 tenths, 9 extra squares = 9 hundredths.

8 **7** The thousandths digit of 4.317 is 7.

9 **18 and 19** 18.93 is 18 wholes and a part less than one, so it is between 18 and 19.

10 **100** tenths vs thousandths: the digit is 2 places to the left, so 10 × 10 = 100 times as much.